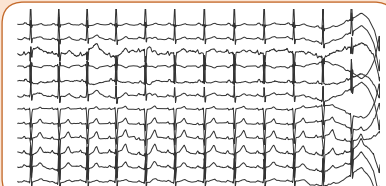


A. Clinical ECG sources

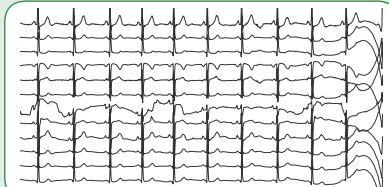
PTB-XL



Georgia



Chapman



Ningbo



Harmonised label space

13 diagnostic labels retained
across all sources | 66,379 recordings

Leave-one-source-out evaluation

Fold 1

Test:

PTB-XL

Fold 2

Test:

Georgia

Fold 3

Test:

Chapman

Fold 4

Test:

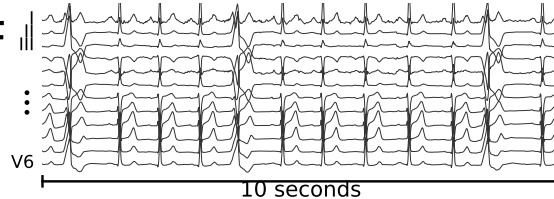
Ningbo

train on three sources, test on the fourth
(four rotations)

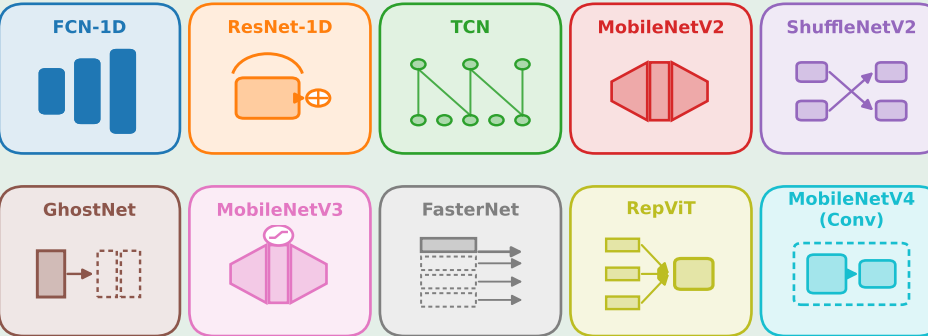
B. Lightweight architecture benchmark

12-lead ECG input

10 s at 100 Hz
(12 x 1,000 samples)



10 encoder families × 2 size budgets



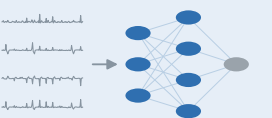
Compact

≤ 32 KB INT8 weights

Standard

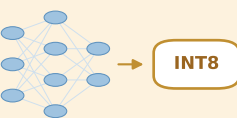
≤ 128 KB INT8 weights

Train (FP32)



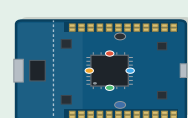
same protocol for
all models and sources

Quantise (INT8)



post-training
quantisation

Deploy (STM32F411)

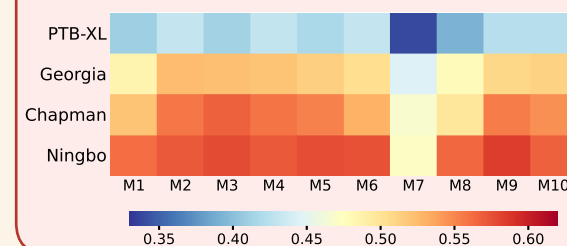


measured on device:
Flash, SRAM, latency

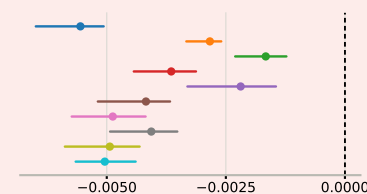
C. Evaluation outputs

RQ1. External discrimination

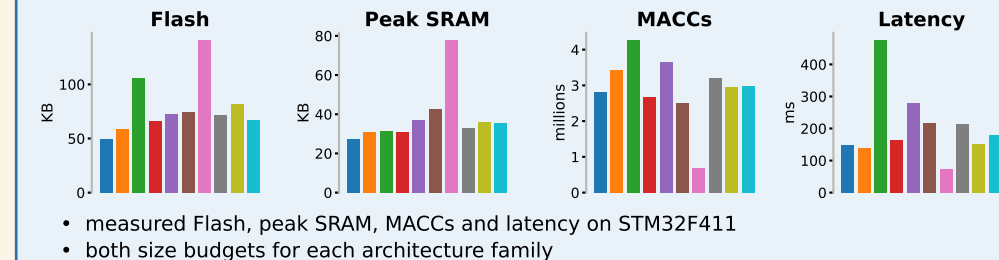
macro-AUPRC (FP32)



INT8 - FP32 (equal-source)



RQ2. Hardware characterisation



RQ3. Performance-resource trade-offs

